Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit

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Abstract

We call a two-time family of transition kernels on a finite state space flat when it satisfies the Chapman–Kolmogorov property, and we read flatness as a flat transport over the complex of time intervals. This paper develops the geometry of families for which flatness fails; such a family admits no time-local Markov generator on the observed state space. The defect $\Delta(t, s, r)$ is the intrinsic failure of the Chapman–Kolmogorov property; it is a kernel-valued 2-cochain, and associativity of composition imposes on it a coherence identity of Bianchi type. A first result closes the natural route to an invariant: under the invertible mass-preserving re-dressings admissible for a family, the orbit closure of every family contains flat families, so no orbital minimization of a flatness distance defines an invariant. The invariant is built from dilations instead. Within a declared dilation class, $\mathcal{F}_d$ denotes the compact semialgebraic closure of the set of compressions of flat transports on d hidden states with positive block masses. The liftability profile $h_d(V)$, the attained distance from V to $\mathcal{F}_d$ in a weighted metric on families, is nonincreasing in d and 1-Lipschitz in the family, and its zero stratum consists of the families approximable to arbitrary precision by such compressions. On a grid of $m+1$ times with n observed states the model sets stabilize: every conditional family of a process on the grid lies in $\mathcal{F}_D$ for an explicit $D \le D_{\max} = \sum_{k=0}^{m} n^{k+1}$, so $\mathcal{F}_d = \mathcal{F}_{D_{\max}}$ for every $d \ge D_{\max}$, the union of the model sets is closed, and the profile of every family in the closure of the process-induced families vanishes at a dimension at most $D_{\max}$. The memory rank, the minimal vanishing dimension, is thus finite exactly on that closure. Two linear programs bound the profile from below: the marginal-consistency defect and the joint-law defect; the second vanishes on the closure of the process-induced families, characterizes the process-induced families through the support of its optimal laws, and detects families that the first does not. First-order conditions at the projection are stated on the tangent and normal cones of the strata. The profile is class-relative: passing from time-homogeneous to time-inhomogeneous lifts weakly decreases it and decreases it strictly on explicit families at the dimension-matched level. Exact examples close the paper: a family whose profile is bounded below at every dimension by a direct evaluation of the obstructions, a family with consistent marginals and no joint law, a lumped chain whose memory rank lies strictly below its generating dimension, and the class gap in closed form.

Keywords. Chapman–Kolmogorov divisibility; indivisible stochastic processes; hidden Markov models; semialgebraic sets; metric algebraic geometry; realization theory; memory rank.

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1 Introduction

Let $\{V(t \leftarrow s)\}_{s \le t}$ be a two-time family of column-stochastic transition kernels on a finite state space, indexed by a finite grid of times. If the family arises as the conditional transition probabilities of a Markov chain observed in full, it composes:

$$V(t \leftarrow s) \;=\; V(t \leftarrow r)\, V(r \leftarrow s), \qquad s \le r \le t, \tag{1}$$

which is the Chapman–Kolmogorov property. We read (1) geometrically: V defines a flat transport over the complex of time intervals. If instead the family arises by observing a Markov system through a lumping of its states, the property (1) generically fails [20]. The failure is quantified by the defect

$$\Delta_V(t, s, r) \;=\; V(t \leftarrow s) - V(t \leftarrow r)\, V(r \leftarrow s), \qquad s < r < t, \tag{2}$$

the object called the indivisibility deficit in the program of which this paper is a part [27, 28, 29, 30]. The quantum-information literature on non-Markovian dynamics measures the failure of divisibility of a dynamical map by analogous quantities [25, 6, 24]; the correspondence is formal, with column-stochastic kernels in place of completely positive maps and the deficit (2) in place of the divisibility defect, and we use it as an analogy only. Proposition 3.6 of [28] shows that a family violating (1) admits no time-ordered-exponential generator on its own state space in the strongly measurable, Bochner-integrable class; on a finite grid the point is elementary, since the two-time evolution of any Markov semigroup on the observed states, time-dependent or otherwise, composes. The phrase "without a generator" in the title carries this meaning throughout; a generator on a larger hidden state space may exist, and the dilations of Section 4 are exactly such generators. The deficit is thus an intrinsic property of the observed family, in contrast with the germ defects of sewing theory [15, 12, 22, 7, 23], which measure a local approximation scheme. The paper converts Δ_V into a well-defined, stable invariant, computable in principle and bounded from below by the linear programs of Section 5.

Two alternatives to the construction fail, and the paper records the failures as theorems because they close routes a reader will otherwise propose. First, the deficit cannot be normalized away: re-dressings of the family by invertible mass-preserving changes of description transform Δ_V by invertible two-sided multiplication and never annihilate it (Lemma 3.2). Second, the deficit cannot be minimized over such re-dressings either: Theorem 3.3 shows that the closure of the orbit of *every* family under the re-dressings admissible for it contains flat families, so the orbital infimum of any continuous flatness-distance is identically zero and carries no information. The mechanism is a one-parameter collapse onto rank-one reset kernels. Remark 3.6 identifies the re-dressing group with the similarity group of quasi-realizations in hidden-Markov realization theory, so the theorem is a statement about a recognized object. The gauge route to an invariant is empty within that class.

The invariant is a metric projection. Fix a dilation class: a class of hidden Markov lifts, each consisting of an initial law and a flat transport on d hidden states together with a labeling surjection onto the observed states, and let $\mathcal{F}_d$ be the closure of the set of observable compressions of such lifts with positive block masses (Definition 4.2). The set $\mathcal{F}_d$ is compact and semialgebraic (Propositions 4.3 and 4.5), the sets are nested in d (Proposition 4.4), and the liftability profile

$$h_d(V) \;=\; \operatorname{dist}\big(V, \mathcal{F}_d\big) \tag{3}$$

is attained (Proposition 4.3), nonincreasing in d (Proposition 4.4), and 1-Lipschitz in the family (Proposition 4.6). Because $\mathcal{F}_d$ is defined as a closure, its zero stratum is the set of families liftable in the closed model class, the families approximable to arbitrary precision by compressions of positive-block-mass d-state lifts (Remark 4.9), and its minimal vanishing dimension defines

the memory rank $\mathrm{rk}_{\mathrm{mem}}(V)$ (Definition 5.1), a closed-model, class-relative integer. Theorem 5.7 gives the realization theory. Every conditional family of a process on the grid lies in some $\kappa(\Theta_D^+)$ exactly, at a dimension D bounded by $D_{\max} = \sum_{k=0}^{m} n^{k+1}$; the model sets stabilize, $\mathcal{F}_d = \mathcal{F}_{D_{\max}}$ for all $d \geq D_{\max}$, so their union is closed and equals the closure of the process-induced families; and the profile of every family in that closure vanishes by dimension $D_{\max}$. The memory rank is thus finite exactly on the closure of the process-induced families, and the asymptotic liftability class, the families whose profile tends to zero without vanishing, is empty (Corollary 5.9). Two linear programs bound the profile from below on families outside the closure: the marginal-consistency defect (Theorem 5.6) and the joint-law defect (Theorem 5.11), the second of which vanishes on the closure of the process-induced families, characterizes the process-induced families through the support of its optimal laws, and is the stronger detector. First-order conditions at the projection are stated on the tangent and normal cones of the strata (Section 6), and the profile is class-relative in a quantifiable way: enlarging the class from time-homogeneous to time-inhomogeneous lifts weakly decreases it and decreases it strictly on explicit families at the dimension-matched level (Proposition 7.1). Section 8 gives exact examples: a family whose profile is bounded below at every dimension by a direct evaluation of both obstructions, a family with consistent marginals and no joint law, a lumped chain whose memory rank lies strictly below its generating dimension, the class gap of Proposition 7.1 in closed form, and the rate of collapse along the gauge path.

The paper states its results in two registers. Several statements of Sections 4 and 6 are standard consequences of the definitions: compactness and attainment follow from the closure in Definition 4.2, stability is the 1-Lipschitz property of every distance function, semialgebraicity is the Tarski–Seidenberg theorem, and the first-order conditions of Section 6 are the general necessary conditions of variational analysis on a closed semialgebraic set. The paper states them because the companion papers use the profile and its hypotheses must be on record. The results proper to this paper are the gauge trivialization theorem with its corollary (Theorem 3.3, Corollary 3.4); the stabilization of the model sets at $D_{\max}$, with the finiteness of the memory rank on the closure of the process-induced families (Theorem 5.7(d), Corollary 5.9); the two linear-programming obstructions with their Lipschitz constants (Theorems 5.6 and 5.11); the characterization of the closure gap and its intrinsic witness (Remark 4.9, Example 4.10); grid monotonicity (Proposition 4.11); and class dependence with an explicit gap (Proposition 7.1).

The paper's tools are those of metric algebraic geometry [5]: the model sets $\mathcal{F}_d$ are compact semialgebraic images of parameter spaces, the invariant is a Euclidean distance to a model set, its smooth critical-point structure is bounded, under standard hypotheses, by the Euclidean distance degree [11], and its regularity theory lives on Whitney stratifications of semialgebraic sets [4] through the normal cones of variational analysis [26]. The memory rank refines a classical circle of ideas, and the profile has neighbors in several literatures. In the linear setting, the minimal dimension of a realization of an input-output map is a Hankel rank, computed by the Ho–Kalman construction [17], and the approximate minimization of weighted automata in that setting is solved exactly by Adamyan–Arov–Krein theory [3]; the model set there is a linear-algebraic object, while positivity of the lift makes $\mathcal{F}_d$ semialgebraic and nonconvex, which is the obstruction that extension-complexity theory describes when lifts of one object through different cones have different minimal dimensions [33]. For hidden Markov models the realization problem is long-studied and known to be delicate for the same reason [16, 1, 31], the identifiability and minimal-dimension questions have an information-theoretic form [18], the model varieties have an algebraic-statistical description [10], and the order of a hidden Markov model is estimated from data with known error exponents [14]. Two constructions are the closest relatives of the profile. Finesso, Grassi, and Spreij [13] approximate a stationary process by hidden Markov models of assigned order in an information divergence rate on the full process; the profile instead measures a Euclidean distance on two-time conditionals to a model set indexed by hidden dimension, on a finite grid, with the realization statements of

Theorem 5.7 and the two obstructions of Section 5 as its added content. Casanellas, Fernández-Sánchez, and Garrote-López [8] compute distances to the stochastic part of a model variety; h_d is the temporal instance of that computation, and the semialgebraic boundary of the nonnegative-rank sets studied by Kubjas, Robeva, and Sturmfels [21], with the nonnegative rank of Cohen and Rothblum [9], is the one-time analog of the boundary of $\mathcal{F}_d$ that Remark 4.9 describes. The joint-law defect of Section 5.3 is a finite marginal problem in the sense of Vorob'ev and Kellerer [32, 19], and the compatibility of a family of conditionals with a joint law is the subject of [2]. Proposition 5.3 calibrates the profile against the deficit norms of [28, 30] through a two-sided Łojasiewicz comparison.

The contributions, in the order of the paper: (i) the coherence identity of Bianchi type that associativity forces on the deficit, with the curvature formula for the twisted coboundary that makes the analogy exact (Proposition 2.2, Remark 2.3); (ii) the gauge trivialization theorem, its corollary, and the identification of the re-dressing group (Theorem 3.3, Remark 3.6); (iii) the construction of the liftability profile with compactness, attainment, monotonicity, semialgebraicity, stability, weight independence of the zero sets, the closed-model reading of the zero stratum with its intrinsic witness, and grid monotonicity (Section 4); (iv) memory rank, the realization theorem with stabilization at $D_{\max}$, and the two linear-programming obstructions (Section 5); (v) first-order conditions at the projection on stratified model sets (Section 6); (vi) class dependence with a quantitative example (Section 7); and (vii) exact examples of the obstructions, of the rank gap, of the class gap, and of the gauge path (Section 8). Section 9 states open problems, including the bridge to the selection functionals of [30] and the compactification question that the closed-model reading opens.

2 Transition families and the deficit

2.1 Families, metric, and flatness

Fix an integer $n \geq 2$, a finite time grid $\mathbb{T} = \{t_0 < t_1 < \cdots < t_m\}$ with $m \geq 2$, and write i for t_i when no confusion arises. Let $\mathcal{S}_n$ denote the set of column-stochastic $n \times n$ matrices, $A \geq 0$ entrywise and $\mathbf{1}^\top A = \mathbf{1}^\top$. A *transition family* is an element

$$V = \{V(j \leftarrow i)\}_{0 \leq i < j \leq m} \ \in \ \mathcal{V} := \prod_{i<j} \mathcal{S}_n.$$

The convention matches [28, 30], where the family is written $\Gamma(t \leftarrow s)$; we write V throughout. We adopt the convention $V(i \leftarrow i) = I$, so the degenerate instances $r \in \{s, t\}$ of (1) are trivial; the same convention $W_\theta(i \leftarrow i) = I$ applies to the transports of Section 4. The space $\mathcal{V}$ is a compact convex subset of the affine space $\mathcal{V}^{\mathrm{aff}}$ of families of real matrices with unit column sums, itself a subset of the vector space of all families of real $n \times n$ matrices.

Fix once and for all strictly positive reference weights: numbers $p_{ij} > 0$ for $i < j$ and vectors $\omega_i \in \mathbb{R}^n_{>0}$. Applications may take them from the law of an observed process, for instance ω_i the occupation law at time t_i; any positive choice is admissible, and the weights are held fixed thereafter. They induce the inner product and norm

$$\langle A, B \rangle \ = \ \sum_{i<j} p_{ij} \sum_{a,b=1}^{n} \omega_i(a)\, A(j \leftarrow i)_{ba}\, B(j \leftarrow i)_{ba}, \qquad \|A\| = \langle A, A \rangle^{1/2}, \tag{4}$$

on the ambient space of kernel families, and the metric $\mathrm{dist}(V, W) = \|V - W\|$ on $\mathcal{V}$. All norms on the ambient space are equivalent; Lemma 4.8 records that the zero sets and the rank do not depend on the choice of weights.

Definition 2.1. $V \in \mathcal{V}$ is *flat* if (1) holds for all $i \le k \le j$ in the grid. Equivalently, $V(j \leftarrow i) = M_j M_{j-1} \cdots M_{i+1}$ where $M_k := V(k \leftarrow k-1) \in \mathcal{S}_n$. The set of flat families is denoted $\mathcal{F}\mathrm{lat} \subset \mathcal{V}$; it is the zero set of the polynomial system $\Delta_V = 0$ in $\mathcal{V}$, so it is closed, and it is compact as a closed subset of $\mathcal{V}$.

2.2 The deficit as a 2-cochain and its coherence identity

For $s < r < t$ in the grid define $\Delta_V(t,s,r)$ by (2). Since $\mathbf{1}^\top V = \mathbf{1}^\top$ for every kernel, $\mathbf{1}^\top \Delta_V(t,s,r) = 0$: the deficit takes values in the space of matrices with vanishing column sums. A kernel-valued k-cochain on the grid assigns a real $n \times n$ matrix to each set of $k+1$ grid points, written with its arguments in the order of (2): the latest time first, then the earliest, then the intermediate times in increasing order. The family V is a 1-cochain on intervals, and $(t,s,r) \mapsto \Delta_V(t,s,r)$ is the 2-cochain that measures its failure to be multiplicative. Associativity of matrix multiplication forces a syzygy.

Proposition 2.2 (Coherence identity)**.** *For all $s < r < q < t$ in the grid,*

$$\Delta_V(t,s,r) \;-\; \Delta_V(t,s,q) \;-\; V(t \leftarrow q)\,\Delta_V(q,s,r) \;+\; \Delta_V(t,r,q)\,V(r \leftarrow s) \;=\; 0. \tag{5}$$

Proof. By (2), $\Delta_V(t,s,r) - \Delta_V(t,s,q) = V(t \leftarrow q)V(q \leftarrow s) - V(t \leftarrow r)V(r \leftarrow s)$. Substitute $V(q \leftarrow s) = \Delta_V(q,s,r) + V(q \leftarrow r)V(r \leftarrow s)$ and $V(t \leftarrow r) = \Delta_V(t,r,q) + V(t \leftarrow q)V(q \leftarrow r)$; the two triple products $V(t \leftarrow q)V(q \leftarrow r)V(r \leftarrow s)$ cancel, leaving $V(t \leftarrow q)\Delta_V(q,s,r) - \Delta_V(t,r,q)V(r \leftarrow s)$. □

Remark 2.3 (The twisted coboundary and its square). Define the V-twisted coboundary on kernel-valued 1-cochains and 2-cochains, for $s < r < t$ and $s < r < q < t$ respectively, by

$$\begin{aligned}(\delta_V \Psi)(t,s,r) &= V(t \leftarrow r)\,\Psi(r,s) - \Psi(t,s) + \Psi(t,r)\,V(r \leftarrow s),\\ (\delta_V \Phi)(t,s,r,q) &= V(t \leftarrow q)\,\Phi(q,s,r) - \Phi(t,s,r) + \Phi(t,s,q) - \Phi(t,r,q)\,V(r \leftarrow s).\end{aligned}$$

Identity (5) is the statement $\delta_V \Delta_V = 0$. When $\delta_V(\delta_V \Psi)$ is expanded from the definitions, eight of the twelve terms cancel in pairs and the remaining four combine into two, which gives for every 1-cochain Ψ

$$(\delta_V^2 \Psi)(t,s,r,q) \;=\; \Psi(t,q)\,\Delta_V(q,s,r) \;-\; \Delta_V(t,r,q)\,\Psi(r,s), \qquad s < r < q < t. \tag{6}$$

So δ_V^2 vanishes when V is flat, and in general δ_V^2 is the action of the deficit on 1-cochains: the pair (5)–(6) is the exact analog of the pair $d_A F_A = 0$, $d_A^2 = F_A$ for a connection A with curvature F_A. This is the whole cohomological content used in the paper. Identity (5) also serves as a correctness check on any implementation that estimates deficits from data.

3 The gauge route and its trivialization

The natural first attempt at an invariant declares part of the deficit to be an artifact of description and removes it by re-dressing. This section defines the re-dressings, proves that they transform the deficit by invertible factors on both sides and never annihilate it, and then proves the sharper negative result: they shrink it, in norm, to zero, for every family, so the orbital infimum of any continuous flatness-distance is identically zero. The construction is explicit, and the conclusion is relative to the class of re-dressings declared in Definition 3.1; Remark 3.6 identifies that class.

Definition 3.1. A *re-dressing* is a tuple $g = (g_0, \dots, g_m)$ of invertible real $n \times n$ matrices with $\mathbf{1}^\top g_i = \mathbf{1}^\top$ for every i. It acts on the affine space $\mathcal{V}^{\mathrm{aff}}$ of families with unit column sums by

$$V^g(j \leftarrow i) \;=\; g_j \, V(j \leftarrow i) \, g_i^{-1}.$$

Since $\mathbf{1}^\top g_j = \mathbf{1}^\top$ implies $\mathbf{1}^\top g_j^{-1} = \mathbf{1}^\top$, column sums are preserved: $\mathbf{1}^\top V^g = \mathbf{1}^\top$. The re-dressing is *admissible for* V if in addition $V^g(j \leftarrow i) \geq 0$ entrywise for all $i < j$, so that $V^g \in \mathcal{V}$. The *orbit* of V is $\mathcal{O}(V) = \{V^g : g \text{ admissible for } V\} \subseteq \mathcal{V}$, the portion of the affine orbit that lies in the stochastic domain.

Lemma 3.2 (Covariance)**.** *For every re-dressing g admissible for V and all $s < r < t$,*

$$\Delta_{V^g}(t, s, r) \;=\; g_t \, \Delta_V(t, s, r) \, g_s^{-1}.$$

In particular, $\Delta_{V^g} = 0$ if and only if $\Delta_V = 0$: no re-dressing annihilates a nonzero deficit, and re-dressings map $\mathcal{F}\mathrm{lat}$ to $\mathcal{F}\mathrm{lat}$.

Proof. $V^g(t \leftarrow s) - V^g(t \leftarrow r)V^g(r \leftarrow s) = g_t V(t \leftarrow s) g_s^{-1} - g_t V(t \leftarrow r) g_r^{-1} g_r V(r \leftarrow s) g_s^{-1} = g_t \Delta_V(t, s, r) g_s^{-1}$, and g_t, g_s are invertible. □

Lemma 3.2 suggests defining the removable deficit as the part that re-dressing can shrink, and the invariant as the orbital minimum of a flatness-distance. Theorem 3.3 shows that this orbital minimum is zero for every family.

Theorem 3.3 (Gauge trivialization)**.** *Let $V \in \mathcal{V}$ be arbitrary and let $w \in \mathbb{R}^n_{>0}$ be any strictly positive probability vector. For $\varepsilon \in (0, 1)$ define*

$$g_i^{(\varepsilon)} \;=\; \varepsilon^{\,i+1} \, I \;+\; \big(1 - \varepsilon^{\,i+1}\big) \, w\mathbf{1}^\top, \qquad i = 0, \dots, m.$$

Then for all sufficiently small ε the re-dressing $g^{(\varepsilon)}$ is admissible for V, and as $\varepsilon \downarrow 0$,

$$V^{g^{(\varepsilon)}}(j \leftarrow i) \;\longrightarrow\; w\mathbf{1}^\top \qquad \text{for every pair } i < j.$$

The limit family, all of whose kernels equal the reset kernel $w\mathbf{1}^\top$, is flat. So the closure of the orbit of every family contains flat families, and

$$\inf_{g \text{ admissible}} \operatorname{dist}\big(V^g, \mathcal{F}\mathrm{lat}\big) \;=\; 0 \qquad \text{for every } V \in \mathcal{V}.$$

Proof. Write $\varepsilon_i = \varepsilon^{\,i+1}$ and $g_i = \varepsilon_i I + (1 - \varepsilon_i) w\mathbf{1}^\top$. Since $\mathbf{1}^\top w = 1$ we have $\mathbf{1}^\top g_i = \mathbf{1}^\top$, and g_i is invertible with

$$g_i^{-1} \;=\; \tfrac{1}{\varepsilon_i}\Big(I - (1 - \varepsilon_i)\, w\mathbf{1}^\top\Big),$$

as direct multiplication verifies using $(w\mathbf{1}^\top)^2 = w\mathbf{1}^\top$. For $i < j$ put $\rho_{ij} = \varepsilon_j / \varepsilon_i = \varepsilon^{\,j-i} \leq \varepsilon$ and $A = V(j \leftarrow i)$. Using $\mathbf{1}^\top A = \mathbf{1}^\top$,

$$g_j A \, g_i^{-1} = \tfrac{1}{\varepsilon_i}\big(\varepsilon_j A + (1 - \varepsilon_j) w\mathbf{1}^\top\big)\big(I - (1 - \varepsilon_i) w\mathbf{1}^\top\big) = \rho_{ij}\, A \;-\; \rho_{ij}(1 - \varepsilon_i)\, Aw\mathbf{1}^\top \;+\; (1 - \varepsilon_j)\, w\mathbf{1}^\top,$$

where the cross terms were simplified with $\mathbf{1}^\top w = 1$ and $\mathbf{1}^\top A = \mathbf{1}^\top$. The (b, a) entry equals $(1 - \varepsilon_j) w_b + \rho_{ij}\big(A_{ba} - (1 - \varepsilon_i)(Aw)_b\big) \;\geq\; (1 - \varepsilon) \min_b w_b - \varepsilon$, which is positive for all pairs simultaneously once $\varepsilon < \min_b w_b / (1 + \min_b w_b)$; so $g^{(\varepsilon)}$ is admissible. Letting $\varepsilon \downarrow 0$ gives $\rho_{ij} \to 0$ and $V^{g^{(\varepsilon)}}(j \leftarrow i) \to w\mathbf{1}^\top$ for every pair. The reset family is flat because $(w\mathbf{1}^\top)(w\mathbf{1}^\top) = w\mathbf{1}^\top$. Since $\operatorname{dist}(\cdot, \mathcal{F}\mathrm{lat})$ is continuous and vanishes at the limit, the infimum over the orbit is 0. □

Corollary 3.4. *Let $F : \mathcal{V} \to [0, \infty)$ be continuous, constant on admissible orbits, and zero on $\mathcal{F}\mathrm{lat}$. Then $F \equiv 0$.*

Proof. F is constant on each orbit, so on its closure by continuity; by Theorem 3.3 the closure contains a flat family, where F vanishes. □

Remark 3.5 (Orbit degeneration). The re-dressing group is the subgroup of $\mathrm{GL}_n(\mathbb{R})^{m+1}$ fixing the covector $\mathbf{1}^\top$ at each time, and $t \mapsto g^{(\varepsilon)}$ with $\varepsilon = e^{-t}$ is a one-parameter subgroup along which every family is driven to the reset locus. The picture is an analogy with orbit degeneration in invariant theory, where a one-parameter subgroup drives a point to a closed orbit in the closure of its own; we use the analogy as a description and prove nothing in that language. Corollary 3.4 is the proved statement: on the admissible stochastic domain, every continuous functional that is constant on orbits and vanishes on $\mathcal{F}$lat vanishes identically. We claim nothing about a quotient or about rings of polynomial invariants for the full group action. Replacing the invariant by a bounded-conditioning surrogate does not help: Remark 3.6 shows that the surrogate is not constant on orbits and that it collapses as the bound relaxes. The remedy of the next section abandons the quotient for a metric projection onto a fixed model set, a construction to which orbit degeneration is irrelevant.

Remark 3.6 (The re-dressing group in realization theory; the two extremes). The re-dressings of Definition 3.1 are the similarity transformations of hidden-Markov realization theory. A quasi-realization of a finite-valued process consists of a row vector, a family of matrices indexed by the symbols, and a column vector reproducing the string probabilities; two quasi-realizations of the same dimension related by an invertible similarity describe the same process; fixing the mass covector is the normalization that keeps the terminal vector equal to $\mathbf{1}$; and a quasi-realization of dimension D yields a positive realization of the same dimension exactly when a similarity maps its parameters into the nonnegative cone, while a minimal positive realization may require more states than a minimal quasi-realization [1, 31]. Definition 3.1 applies this similarity at each grid time to the two-time kernels, and admissibility is the cone condition. So Theorem 3.3 concerns a recognized group, and its interaction with the non-uniqueness results of positive realization theory is a natural question that we record without answering. The class also has two extremes. A stochastic matrix with a stochastic inverse is a permutation matrix. Let g be a re-dressing such that g and g^{-1} both map $\mathcal{V}$ into itself. Applied to the flat family whose kernels all equal the reset kernel $e_b\mathbf{1}^\top$, with e_b the b-th standard basis vector, the action gives $g_j e_b \mathbf{1}^\top g_i^{-1} = g_j e_b \mathbf{1}^\top \geq 0$ for every $j \geq 1$ and every b, so $g_1, \ldots, g_m$ are entrywise nonnegative, and the same argument for g^{-1} gives $g_1^{-1}, \ldots, g_m^{-1} \geq 0$; so $g_1, \ldots, g_m$ are permutations. Applied to the flat family whose kernels all equal I, the action gives $g_j g_0^{-1} \geq 0$ and $g_j^{-1} g_0 \geq 0$, so $g_0^{-1} = g_j^{-1}(g_j g_0^{-1})$ and $g_0 = g_j(g_j^{-1} g_0)$ are nonnegative, and g_0 is a permutation as well. Thus g is a time-dependent relabeling. With uniform weights the deficit norm is invariant under time-dependent relabelings, so the orbital minimum over this subclass is the deficit norm itself. The full class of Definition 3.1 collapses every orbit onto $\mathcal{F}$lat by Theorem 3.3, so the orbital infimum over it is zero for every family. An intermediate class, the re-dressings with $\|g_i\|_{\mathrm{op}} \|g_i^{-1}\|_{\mathrm{op}} \leq K$ for every i and a fixed $K \geq 1$, yields no invariant either. Let $V \notin \mathcal{F}$lat and let $F_K(V)$ be the infimum of $\mathrm{dist}(V^g, \mathcal{F}\mathrm{lat})$ over the re-dressings of the class admissible for V. Since $\mathbf{1}^\top g_i = \mathbf{1}^\top$ forces $\|g_i\|_{\mathrm{op}} \geq 1$ and $\|g_i^{-1}\|_{\mathrm{op}} \geq 1$, the class is bounded; it is closed, because a limit of matrices with $\|g^{-1}\|_{\mathrm{op}} \leq K$ is invertible with the same bound, and inversion is continuous on it; admissibility is a closed condition on it; and $V^g \notin \mathcal{F}$lat for every admissible g by Lemma 3.2. So $F_K(V)$ is attained and positive. Along the path of Theorem 3.3 the families $V^{g^{(\varepsilon)}}$ lie on the orbit of V, and the identity re-dressing, of conditioning $1 \leq K$, is admissible for each of them, so $F_K(V^{g^{(\varepsilon)}}) \leq \mathrm{dist}(V^{g^{(\varepsilon)}}, \mathcal{F}\mathrm{lat}) \to 0$. Thus F_K is not constant on orbits, and Example 8.5 gives the rate at which it tends to zero as $K \to \infty$. The conclusion of this section is thus relative to the declared class.

4 Dilation classes and the liftability profile

4.1 Lifts, compressions, and the model sets

Definition 4.1 (Dilation class and lifts)**.** Fix $d \geq n$. A *d-state flat lift* over the grid is a triple $\theta = (\pi, (M_k)_{k=1}^m, \varphi)$ where π is a probability vector on $[d]$, each M_k is a column-stochastic $d \times d$ matrix, the transport upstairs is $W_\theta(j \leftarrow i) = M_j M_{j-1} \cdots M_{i+1}$, and $\varphi : [d] \to [n]$ is a surjection, the *labeling*. The *time-homogeneous class* $\mathfrak{D}^{\mathrm{hom}}$ requires $M_1 = \cdots = M_m$; the *time-inhomogeneous class* $\mathfrak{D}^{\mathrm{inhom}}$ imposes no such constraint. Write $p_i(\theta) = W_\theta(i \leftarrow 0)\pi \in \mathbb{R}^d$ for the hidden law at time t_i and $\mu_i(\theta) \in \mathbb{R}^n$, $\mu_i(\theta)_a = \sum_{u:\varphi(u)=a} p_i(\theta)_u$, for the observed law. Let Θ_d^+ be the set of lifts in the declared class with $\mu_i(\theta)_a > 0$ for all i, a; it is the finite disjoint union, over the labelings φ, of the sets $\Theta_{d,\varphi}^+$ of lifts with that labeling. We call the classes *dilation classes*.

Definition 4.2 (Compression and model sets)**.** For $\theta \in \Theta_d^+$ the *compression* $\kappa(\theta) \in \mathcal{V}$ is the family of block-conditional kernels of the lifted chain:

$$\kappa(\theta)(j \leftarrow i)_{ba} \;=\; \mathbb{P}_\theta\big(\varphi(X_{t_j}) = b \,\big|\, \varphi(X_{t_i}) = a\big) \;=\; \frac{\mathbf{1}_b^\top W_\theta(j \leftarrow i)\,\mathrm{diag}(\mathbf{1}_a)\,p_i(\theta)}{\mu_i(\theta)_a}, \tag{7}$$

where $\mathbf{1}_a \in \{0,1\}^d$ indicates the block $\varphi^{-1}(a)$ and (X_{t_i}) is the hidden chain with initial law π. The *model set* in the declared class is

$$\mathcal{F}_d \;:=\; \mathrm{cl}\big(\kappa(\Theta_d^+)\big) \;=\; \bigcup_{\varphi} \mathrm{cl}\big(\kappa(\Theta_{d,\varphi}^+)\big) \subseteq \mathcal{V},$$

the union being over the finitely many labelings, and the *liftability profile* of $V \in \mathcal{V}$ is $h_d(V) = \mathrm{dist}(V, \mathcal{F}_d)$ as in (3), computed in the metric (4). When the class must be displayed we write $\mathcal{F}_d^{\mathrm{hom}}$, h_d^{hom}, and likewise for inhom. The set $\mathcal{F}_d$ is a closure: membership of V in $\mathcal{F}_d$ asserts approximability of V, to arbitrary precision, by compressions of positive-block-mass d-state lifts, and does not by itself assert the existence of an exact lift of V (Remark 4.9, Example 4.10).

The compression (7) is the transition family observed when a Markov system is watched through the lumping φ; families of this form are the classical source of Chapman–Kolmogorov violations [20]. The closure in Definition 4.2 adjoins the families reached only through degenerating block masses; Remark 4.9 characterizes them and Example 4.10 exhibits one.

Proposition 4.3 (Compactness and attainment)**.** *$\mathcal{F}_d$ is a nonempty compact subset of $\mathcal{V}$, and for every $V \in \mathcal{V}$ the distance $h_d(V)$ is attained at some $W^* \in \mathcal{F}_d$.*

Proof. $\mathcal{F}_d \subseteq \mathcal{V}$ is bounded, and it is closed by construction; a closed subset of the compact set $\mathcal{V}$ is compact. Nonemptiness: the lift with uniform π, every M_k equal to the uniform reset kernel $d^{-1}\mathbf{1}\mathbf{1}^\top$, and any surjective labeling has block masses $|\varphi^{-1}(a)|/d > 0$ at every time, so it lies in Θ_d^+. Distance to a nonempty compact set is attained. □

Proposition 4.4 (Nesting and monotonicity)**.** *In either class, $\kappa(\Theta_d^+) \subseteq \kappa(\Theta_{d+1}^+)$ and $\mathcal{F}_d \subseteq \mathcal{F}_{d+1}$ for all $d \geq n$; so $h_{d+1}(V) \leq h_d(V)$ for every V.*

Proof. Let $\theta = (\pi, (M_k), \varphi) \in \Theta_d^+$ and fix a hidden state u to be split into two copies $u', u'' \in [d+1]$. Let $\psi : [d+1] \to [d]$ be the merge map that sends u' and u'' to u and is a bijection elsewhere. Define the *merging matrix* $P \in \{0,1\}^{d \times (d+1)}$ by $P_{vx} = 1$ exactly when $\psi(x) = v$, and the *splitting matrix* $R \in \mathbb{R}_{\geq 0}^{(d+1)\times d}$ by $R_{xv} = \frac{1}{2}$ if $v = u$ and $x \in \{u', u''\}$, $R_{xv} = 1$ if $v \neq u$

and $\psi(x) = v$, and $R_{xv} = 0$ otherwise. Each column of R is a probability vector supported on $\psi^{-1}(v)$, each column of P has exactly one unit entry, and

$$PR = I_d, \qquad \mathbf{1}_{d+1}^\top R = \mathbf{1}_d^\top, \qquad \mathbf{1}_d^\top P = \mathbf{1}_{d+1}^\top.$$

Define the split lift $\theta' = (\pi', (M'_k), \varphi')$ on $[d+1]$ by $\pi' = R\pi$, $M'_k = RM_kP$, $\varphi' = \varphi \circ \psi$. Then π' is a probability vector, $\mathbf{1}_{d+1}^\top M'_k = \mathbf{1}_d^\top M_k P = \mathbf{1}_d^\top P = \mathbf{1}_{d+1}^\top$, so each M'_k is column-stochastic, and the same P, R at every step preserve homogeneity, so the construction applies verbatim in $\mathfrak{D}^{\mathrm{hom}}$. The two copies of u have identical columns in M'_k, so the merge map satisfies the strong lumpability criterion of Kemeny and Snell [20], in matrix form $PM'_k = M_kP$.

By $PR = I_d$, $M'_kM'_{k-1} = RM_kM_{k-1}P$ and inductively $W_{\theta'}(j \leftarrow i) = R\,W_\theta(j \leftarrow i)\,P$, so $p_i(\theta') = R\,p_i(\theta)$. The block indicators satisfy $\mathbf{1}'^\top_a R = \mathbf{1}_a^\top$, because column v of R is supported on $\psi^{-1}(v)$, on which φ' is constant with value $\varphi(v)$, and sums to 1; by the same support property, $P\,\mathrm{diag}(\mathbf{1}'_a)\,R = \mathrm{diag}(\mathbf{1}_a)$. So $\mu_i(\theta')_a = \mathbf{1}'^\top_a R\,p_i(\theta) = \mu_i(\theta)_a > 0$, and the block conditionals (7) agree:

$$\begin{aligned}\mathbf{1}'^\top_b\, W_{\theta'}(j \leftarrow i)\,\mathrm{diag}(\mathbf{1}'_a)\,p_i(\theta') &= \mathbf{1}_b^\top\, W_\theta(j \leftarrow i)\,\big(P\,\mathrm{diag}(\mathbf{1}'_a)\,R\big)\,p_i(\theta)\\ &= \mathbf{1}_b^\top\, W_\theta(j \leftarrow i)\,\mathrm{diag}(\mathbf{1}_a)\,p_i(\theta).\end{aligned}$$

Thus $\kappa(\theta') = \kappa(\theta)$ with $\theta' \in \Theta^+_{d+1}$, so $\kappa(\Theta^+_d) \subseteq \kappa(\Theta^+_{d+1})$, and closures preserve the inclusion. □

4.2 Semialgebraicity, stability, and weight independence

Proposition 4.5 (Semialgebraicity and stratification)**.** *$\mathcal{F}_d$ is a compact semialgebraic subset of $\mathcal{V}$, and $V \mapsto h_d(V)^2$ is a continuous semialgebraic function. In particular, $\mathcal{F}_d$ admits a Whitney stratification into finitely many smooth semialgebraic strata.*

Proof. Each $\Theta^+_{d,\varphi}$ is semialgebraic (polynomial equalities and strict inequalities), and each entry of $\kappa(\theta)$ in (7) is a rational function of θ whose denominator $\mu_i(\theta)_a$ is positive on $\Theta^+_{d,\varphi}$; the graph of κ is semialgebraic, so its image is semialgebraic by the Tarski–Seidenberg theorem, the closure of a semialgebraic set is semialgebraic, and a finite union over labelings preserves the property [4, Ch. 2]. The squared distance $h_d(V)^2 = \min_{W \in \mathcal{F}_d} \|V - W\|^2$ is semialgebraic as a partial minimization of a polynomial over a semialgebraic set [4, Ch. 2], and its square root h_d is 1-Lipschitz by Proposition 4.6, so $h_d(V)^2$ is continuous. Existence of Whitney stratifications for semialgebraic sets is classical [4, Ch. 9]. □

Proposition 4.6 (Stability)**.** *For all $V, V' \in \mathcal{V}$ and all d, $\big|h_d(V) - h_d(V')\big| \le \mathrm{dist}(V, V')$.*

Proof. Distance functions to a fixed set are 1-Lipschitz: for any $W \in \mathcal{F}_d$, $h_d(V) \le \|V - W\| \le \|V - V'\| + \|V' - W\|$; take the infimum over W and symmetrize. □

Remark 4.7 (Plug-in estimation and a consistent rank rule). Proposition 4.6 transfers norm convergence of a family estimator to the profile: if $\widehat{V}_N$ satisfies $\mathrm{dist}(\widehat{V}_N, V) \to 0$ in some mode (in probability, almost surely, or with a rate), then $h_d(\widehat{V}_N) \to h_d(V)$ in the same mode and with the same rate. It says nothing about distributional limits, which would require a delta-method analysis at points where the distance function is differentiable, and nothing about the integer $\mathrm{rk}_{\mathrm{mem}}(\widehat{V}_N)$, which need not converge to $\mathrm{rk}_{\mathrm{mem}}(V)$: the plug-in rank without a threshold satisfies $\liminf_N \mathrm{rk}_{\mathrm{mem}}(\widehat{V}_N) \ge \mathrm{rk}_{\mathrm{mem}}(V)$, because the model sets are closed and nested, and, when $r = \mathrm{rk}_{\mathrm{mem}}(V) < \infty$, $\mathrm{rk}_{\mathrm{mem}}(\widehat{V}_N) > r$ for every N with $\widehat{V}_N \notin \mathcal{F}_r$; so it is biased upward. A threshold rule is consistent. Suppose $\mathrm{dist}(\widehat{V}_N, V) \le \varepsilon_N$ with $\varepsilon_N \to 0$, let $\tau_N \to 0$ with $\tau_N/\varepsilon_N \to \infty$, and define $\hat{d}_N = \min\{d \ge n : h_d(\widehat{V}_N) \le \tau_N\}$, with $\hat{d}_N = \infty$ when the set is empty. If $r = \mathrm{rk}_{\mathrm{mem}}(V) < \infty$, then $h_r(V) = 0$ and, when $r > n$, $\gamma := h_{r-1}(V) > 0$ by Definition 5.1;

for all large N, $\varepsilon_N \le \tau_N < \gamma - \varepsilon_N$, so $h_r(\widehat{V}_N) \le \varepsilon_N \le \tau_N$ and $h_d(\widehat{V}_N) \ge \gamma - \varepsilon_N > \tau_N$ for every $d < r$ by Propositions 4.4 and 4.6, and $\hat{d}_N = r$. If $\mathrm{rk}_{\mathrm{mem}}(V) = \infty$, then $h_d(V) \ge h_{D_{\max}}(V) > 0$ for every d by Theorem 5.7(d), and the same argument gives $\hat{d}_N = \infty$ for all large N. The margin γ exists but is unknown, so the rate ε_N of the family estimator is the quantity that the threshold must dominate; the statement is the consistency of the integer under the stated hypotheses and no more.

Lemma 4.8 (Weight independence of the zero sets and the rank)**.** *The sets $\{V : h_d(V) = 0\} = \mathcal{F}_d$, and hence the memory rank of Definition 5.1, do not depend on the choice of weights in (4). Only the numerical values $h_d(V) > 0$ do, and any two admissible weight choices yield profiles equivalent up to two-sided multiplicative constants.*

Proof. All norms induced by (4) with strictly positive weights are equivalent on the finite-dimensional ambient space; equivalent norms have identical zero-distance sets and comparable distances. □

4.3 The closure gap and grid refinement

Remark 4.9 (What zero distance means; the closure gap). Since $\mathcal{F}_d = \mathrm{cl}\,\kappa(\Theta_d^+)$, the statement $h_d(V) = 0$ says that V is approximable to arbitrary precision by compressions of positive-block-mass d-state lifts. It does not by itself produce an exact lift of V. The gap $\mathcal{F}_d \setminus \kappa(\Theta_d^+)$ has an exact description: $V \in \mathcal{F}_d$ lies outside $\kappa(\Theta_d^+)$ if and only if every sequence $\theta_\ell \in \Theta_d^+$ with $\kappa(\theta_\ell) \to V$ satisfies $\min_{i,a} \mu_i(\theta_\ell)_a \to 0$. For the nontrivial direction, suppose some approximating sequence has $\min_{i,a} \mu_i(\theta_\ell)_a \ge \eta > 0$ along a subsequence; the parameters $(\pi, (M_k))$ range over a compact set and the labelings are finitely many, so a further subsequence converges to a lift θ^* with block masses at least η, hence $\theta^* \in \Theta_d^+$, and κ is continuous at θ^* because its denominators are bounded below there, so $V = \kappa(\theta^*)$. The gap is thus the set of families reached only through degenerating block masses. At $d = n$ the labelings are bijections, $\kappa(\theta) = W_\theta$ up to relabeling, and the block masses are the entries of the hidden laws $W_\theta(i \leftarrow 0)\pi$. So, in $\mathfrak{D}^{\mathrm{inhom}}$, a flat family V lies in $\kappa(\Theta_n^+)$ if and only if some strictly positive π makes every $V(i \leftarrow 0)\pi$ strictly positive, and the gap at $d = n$ consists exactly of the flat families for which every positive initial law is driven to a law with a zero entry: the flat families with degenerate forced marginals. In $\mathfrak{D}^{\mathrm{hom}}$ the same statements hold with "flat family" replaced by "flat family with equal one-step kernels", since $\mathcal{F}_n^{\mathrm{hom}}$ consists of those (Example 5.2). Example 4.10 exhibits a member of the gap in both classes. For $d > n$, whether a family in the gap can admit strictly positive consistent one-time marginals in the sense of Lemma 5.4 is open. Two readings of the zero stratum are available. The reading adopted throughout this paper, in Definition 5.1 in particular, is the *closed-model* reading: liftability in dimension d means membership in $\mathcal{F}_d$, and all zero-stratum and rank statements are statements about the closed model class. The alternative would enlarge the class to a compactification, lifts carrying boundary data such as conditional kernels on blocks of vanishing mass, together with a proof that the compactified class's compression image is exactly $\mathcal{F}_d$, which would restore exact-realization language for the zero stratum. Constructing such a class is Open Problem (vii).

Example 4.10 (An intrinsic point of the closure gap). Let $n = d = 2$ and $M = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. The matrix M is column-stochastic and idempotent, so the constant family $V \equiv M$ is flat and $h_2(V) = 0$ by Example 5.2. For every probability vector π, $M\pi = (1, 0)^\top$: the second block has mass zero at every time $t \ge t_1$, so V is not the compression of any lift in Θ_2^+, in either class. With $M_\varepsilon = \left(\begin{smallmatrix} 1-\varepsilon & 1-\varepsilon \\ \varepsilon & \varepsilon \end{smallmatrix}\right)$ and $\pi = (\frac{1}{2}, \frac{1}{2})$, the lift $\theta_\varepsilon = (\pi, M_\varepsilon, \mathrm{id})$ lies in Θ_2^+ with block masses $(\frac{1}{2}, \frac{1}{2})$ at t_0 and $(1 - \varepsilon, \varepsilon)$ at every later time, its compression is the constant family M_ε, and $\mathrm{dist}(V, \kappa(\theta_\varepsilon)) = 2\varepsilon\sqrt{|\{i < j\}|}$ in the metric (4) with unit weights, since each pair contributes

$\|M - M_\varepsilon\|_F^2 = 4\varepsilon^2$; on a grid of three points this is $2\sqrt{3}\,\varepsilon$. So $V \in \mathcal{F}_2 \setminus \kappa(\Theta_2^+)$, and the gap is a property of the family.

Proposition 4.11 (Grid monotonicity)**.** *Let $\mathbb{T}' \subseteq \mathbb{T}$ be a sub-grid with at least three points, carrying the weights p_{ij}, ω_i inherited from $\mathbb{T}$, and let $V|_{\mathbb{T}'}$ denote the restriction of $V \in \mathcal{V}_{\mathbb{T}}$ to the pairs in $\mathbb{T}'$. Then, in $\mathfrak{D}^{\mathrm{inhom}}$, and in $\mathfrak{D}^{\mathrm{hom}}$ whenever consecutive points of $\mathbb{T}'$ are separated by a constant number of grid steps,*

$$h_d\big(V|_{\mathbb{T}'}\big) \;\le\; h_d(V) \quad \textit{for every } d \ge n, \qquad \textit{so} \qquad \mathrm{rk}_{\mathrm{mem}}\big(V|_{\mathbb{T}'}\big) \;\le\; \mathrm{rk}_{\mathrm{mem}}(V).$$

Proof. Restriction $V \mapsto V|_{\mathbb{T}'}$ is a coordinate projection that drops the summands of (4) indexed by pairs outside $\mathbb{T}'$, so it is 1-Lipschitz. It maps lifts to lifts: for $\theta = (\pi, (M_k), \varphi) \in \Theta_d^+(\mathbb{T})$ with $\mathbb{T}' = \{t_{i_0} < \cdots < t_{i_{m'}}\}$, the lift θ' on $\mathbb{T}'$ with initial law $\pi' = p_{i_0}(\theta)$, the same labeling φ, and step matrices $M'_\ell = W_\theta(i_\ell \leftarrow i_{\ell-1})$ has $W_{\theta'}(i_\ell \leftarrow i_k) = W_\theta(i_\ell \leftarrow i_k)$ and hidden laws $p_{i_\ell}(\theta') = W_\theta(i_\ell \leftarrow i_0)\, p_{i_0}(\theta) = p_{i_\ell}(\theta)$ at the retained times, so the same positive block masses there, and $\kappa(\theta') = \kappa(\theta)|_{\mathbb{T}'}$ by (7), which depends at the pair (i, j) only on $W_\theta(j \leftarrow i)$, $p_i(\theta)$, and φ. When the index gaps are constant and θ is homogeneous, all M'_ℓ coincide, so θ' is homogeneous. Thus $\kappa(\Theta_d^+(\mathbb{T}))|_{\mathbb{T}'} \subseteq \kappa(\Theta_d^+(\mathbb{T}'))$, and by continuity of the projection $\mathcal{F}_d(\mathbb{T})|_{\mathbb{T}'} \subseteq \mathcal{F}_d(\mathbb{T}')$. For $W^* \in \mathcal{F}_d(\mathbb{T})$ attaining $h_d(V)$, $h_d(V|_{\mathbb{T}'}) \le \mathrm{dist}(V|_{\mathbb{T}'}, W^*|_{\mathbb{T}'}) \le \mathrm{dist}(V, W^*) = h_d(V)$. The rank inequality follows from the zero sets. □

Refining the grid thus cannot decrease the profile or the rank. The dimension bound $D_{\max}$ of Theorem 5.7 grows exponentially in the number of grid steps, so this paper supplies no uniform bound on the ranks along a refining sequence of grids; whether they stabilize is the continuous-grid question, Open Problem (vi).

5 Memory rank, realization, and the realizability gap

Definition 5.1 (Memory rank)**.** $\mathrm{rk}_{\mathrm{mem}}(V) = \min\{\, d \ge n : \; h_d(V) = 0 \,\}$, with $\mathrm{rk}_{\mathrm{mem}}(V) = \infty$ if no such d exists. By Proposition 4.4 the set of vanishing dimensions is upward closed, so the minimum is well defined, and by Lemma 4.8 it is independent of the metric weights. The rank is relative to the declared dilation class, and it is a *closed-model* dimension (Remark 4.9): $\mathrm{rk}_{\mathrm{mem}}(V) = d$ asserts that V is approximable to arbitrary precision by d-state lifts and by no fewer, not that V is exactly realized by one. Theorem 5.7(d) shows that $\mathrm{rk}_{\mathrm{mem}}(V)$ is finite exactly when V lies in the closure of the process-induced families, and then $\mathrm{rk}_{\mathrm{mem}}(V) \le D_{\max}$.

Example 5.2 (The dimension-matched charge is distance to flat). For $d = n$ every labeling is a bijection, compressions of bijectively labeled lifts are the lifted families themselves up to relabeling, and relabelings of flat families are flat; so $\kappa(\Theta_n^+) \subseteq \mathcal{F}\mathrm{lat}$ in either class, and $\mathcal{F}_n \subseteq \mathcal{F}\mathrm{lat}$. In the time-inhomogeneous class the inclusion is an equality:

$$\mathcal{F}_n^{\mathrm{inhom}} = \mathcal{F}\mathrm{lat}, \qquad h_n^{\mathrm{inhom}}(V) \;=\; \mathrm{dist}(V, \mathcal{F}\mathrm{lat}).$$

The positivity requirement enters here. A flat family with a zero row in some $V(i \leftarrow 0)$ is not the compression of any lift in Θ_n^+, because every hidden law $V(i \leftarrow 0)\pi$ then has a zero entry (Remark 4.9, Example 4.10); the equality survives because $\mathcal{F}_n$ is a closure and the flat families with strictly positive one-step kernels, which lie in $\kappa(\Theta_n^+)$ with any positive π, are dense in $\mathcal{F}\mathrm{lat}$: perturb the one-step kernels of a flat family toward the uniform kernel and rebuild the products. In the time-homogeneous class $\mathcal{F}_n^{\mathrm{hom}}$ is the closed subset of $\mathcal{F}\mathrm{lat}$ on which all one-step kernels coincide, and $\mathcal{F}_n^{\mathrm{hom}} \subsetneq \mathcal{F}\mathrm{lat}$ as soon as $m \ge 2$ (Proposition 7.1). The bottom of the inhomogeneous profile is thus the *dimension-matched geometric charge* associated with the Chapman–Kolmogorov failure: a metric distance to the flat model set, and not a norm of the

deficit Δ_V. The two vanish together and control each other only up to a Łojasiewicz exponent (Proposition 5.3); the raw deficit is a quadratic polynomial in the family by (2), while h_n^{inhom} is the distance to the semialgebraic set $\mathcal{F}\mathrm{lat}$ (Proposition 4.5), and whether that distance admits a decomposition under lumpings of the observed states is Open Problem (v).

Proposition 5.3 (Łojasiewicz comparison with the deficit)**.** *There exist constants $L, C > 0$ and an exponent $\alpha \in (0, 1]$, depending only on n, the grid, and the weights, such that for all $V \in \mathcal{V}$*

$$\tfrac{1}{L}\,\|\Delta_V\| \;\le\; \operatorname{dist}(V, \mathcal{F}\mathrm{lat}) \quad \textit{and} \quad \operatorname{dist}(V, \mathcal{F}\mathrm{lat}) \;\le\; C\,\|\Delta_V\|^{\alpha},$$

where $\|\Delta_V\|$ is any fixed weighted ℓ^2 norm of the triple-indexed deficit, for instance $\|\Delta_V\|^2 = \sum_{s<r<t}\|\Delta_V(t,s,r)\|_F^2$; the constants depend on the choice. So h_n^{inhom} and the integrated deficit norms of [28, 30], which are norms of Δ_V integrated over the time variables, are two-sidedly comparable up to a Łojasiewicz exponent: each vanishes exactly when the other does, and each controls the other on the compact $\mathcal{V}$.

Proof. The map $V \mapsto \Delta_V$ is polynomial, hence Lipschitz on the compact $\mathcal{V}$, and vanishes on $\mathcal{F}\mathrm{lat}$; the first inequality follows by evaluating at a nearest flat family. The second is the Łojasiewicz inequality applied to the semialgebraic function $f(V) = \|\Delta_V\|^2$, whose zero set is $\mathcal{F}\mathrm{lat}$: on a compact semialgebraic set, $\operatorname{dist}(V, f^{-1}(0)) \le C|f(V)|^{\alpha/2}$ for some $C > 0$ and $\alpha \in (0, 1]$ [4, Ch. 2]. □

The comparison is the formal bridge between the projection invariant of this paper and the deficit-norm theory of the companion papers. The constant L is explicit, a Lipschitz constant of the polynomial map $V \mapsto \Delta_V$ on $\mathcal{V}$; the exponent α and the constant C come from the Łojasiewicz inequality and are not constructive here.

5.1 Necessity: the marginal-consistency defect

Lemma 5.4. *Every $W \in \mathcal{F}_d$ (either class) admits probability vectors $\mu_0, \dots, \mu_m \in \Sigma^{n-1}$, the probability simplex in $\mathbb{R}^n$, with*

$$W(j \leftarrow i)\,\mu_i \;=\; \mu_j \qquad \textit{for all } i < j. \tag{8}$$

Proof. For $W = \kappa(\theta)$ with $\theta \in \Theta_d^+$, take $\mu_i = \mu_i(\theta)$, the observed laws; (8) is the law of total probability. For $W \in \operatorname{cl}\kappa(\Theta_d^+)$, pick $\kappa(\theta_\ell) \to W$, pass to a subsequence along which the vectors $\mu_i(\theta_\ell)$ converge in the compact simplex, and take limits in (8). □

Definition 5.5. The *marginal-consistency defect* (the *consistency defect* for short) of $V \in \mathcal{V}$ is

$$\kappa_c(V) \;=\; \min_{\mu \in (\Sigma^{n-1})^{m+1}} \; \max_{i<j} \big\| V(j \leftarrow i)\mu_i - \mu_j \big\|_1.$$

The minimum is attained by compactness, and $V \mapsto \kappa_c(V)$ is continuous. Since each residual $V(j \leftarrow i)\mu_i - \mu_j$ is affine in μ, the minimax is a linear program in epigraph form, and κ_c is computable exactly; Example 8.2 evaluates it by hand. The defect tests the pairwise consistency of one-time marginals only; the joint-law defect of Section 5.3 tests the existence of a joint law and is strictly stronger as a detector.

Theorem 5.6 (Realizability gap)**.** *There is a constant $C_* > 0$, depending only on n, the grid, and the weights, namely*

$$C_* \;=\; \sqrt{n}\,\big(\min_{i<j,\,a}\; p_{ij}\,\omega_i(a)\big)^{-1/2},$$

such that for every $V \in \mathcal{V}$ and every $d \geq n$,

$$h_d(V) \;\geq\; \frac{\kappa_c(V)}{C_*}.$$

In particular, whenever $\kappa_c(V) > 0$, the profile $h_d(V)$ is bounded away from zero uniformly in d and $\mathrm{rk}_{\mathrm{mem}}(V) = \infty$: no dilation of any dimension explains the family, exactly or in the limit.

Proof. κ_c is Lipschitz in V with constant C_*. Fix $\mu \in (\Sigma^{n-1})^{m+1}$ and a pair $i < j$, and write $X = (V - W)(j \leftarrow i)$. Since μ_i is a probability vector,

$$\|X\mu_i\|_1 \;\leq\; \max_a \sum_b |X_{ba}| \;\leq\; \sqrt{n}\, \max_a \Big(\sum_b X_{ba}^2\Big)^{1/2} \;\leq\; \sqrt{n}\, \Big(\sum_{a,b} X_{ba}^2\Big)^{1/2},$$

by convexity in the first step and Cauchy–Schwarz on each column sum in the second. The metric (4) bounds the square sum: $\mathrm{dist}(V,W)^2 \;\geq\; p_{ij}\, \min_{a'} \omega_i(a') \sum_{a,b} X_{ba}^2$, so $\|X\mu_i\|_1 \leq C_*\, \mathrm{dist}(V,W)$ with C_* as displayed. Taking the maximum over pairs and minimizing over μ gives $|\kappa_c(V) - \kappa_c(W)| \leq C_*\, \mathrm{dist}(V,W)$. By Lemma 5.4, $\kappa_c \equiv 0$ on $\mathcal{F}_d$; evaluating at a nearest point W^* gives $\kappa_c(V) \leq C_*\, h_d(V)$. □

The hypothesis $\kappa_c(V) > 0$ is an input to be verified by the linear program of Definition 5.5, not a property the theorem predicts. The obstruction is necessary only: compatible one-time marginals do not by themselves produce a joint law, and no converse is claimed; Theorem 5.11 supplies the stronger test.

5.2 Realization of process-induced families

Call $V \in \mathcal{V}$ *process-induced* if there exists a stochastic process $(X_{t_i})_{i=0}^m$ on $[n]$ with $\mathbb{P}(X_{t_i} = a) > 0$ for all i, a and $\mathbb{P}(X_{t_j} = b \mid X_{t_i} = a) = V(j \leftarrow i)_{ba}$ for all $i < j$ and all a, b. Write $\mathcal{P} \subseteq \mathcal{V}$ for the set of process-induced families and

$$D_{\max} \;:=\; \sum_{k=0}^{m} n^{\,k+1} \;=\; \frac{n\,(n^{m+1} - 1)}{n - 1} \tag{9}$$

for the number of observed histories of length at most $m+1$. The construction in part (a) below is the state augmentation by the observed history, a standard device of realization theory; the content of the theorem is in the dimension bookkeeping of (a) and in the stabilization (d) that the bookkeeping yields.

Theorem 5.7 (Realization and stabilization)**.** *The following hold in either dilation class.*

(a) (Exact realization of process-induced families) If $V \in \mathcal{P}$, then $V \in \kappa(\Theta_D^+)$ exactly, not merely in the closure, with

$$D \;:=\; \sum_{k=0}^{m} N_k \;\leq\; D_{\max},$$

where $N_k \leq n^{k+1}$ is the number of observed histories of length $k+1$ with positive probability under a realizing process. Hence $h_D(V) = 0$ and $\mathrm{rk}_{\mathrm{mem}}(V) \leq D$.

(b) (Approximation) Every element of every $\mathcal{F}_d$ is a limit of process-induced families: $\mathcal{F}_d \subseteq \mathrm{cl}\,\mathcal{P}$ for every d.

(c) (Asymptotic characterization) $\lim_{d\to\infty} h_d(V) = 0$ if and only if $V \in \mathrm{cl}\,\mathcal{P}$.

(d) (Stabilization at $D_{\max}$) $\mathcal{F}_d = \mathcal{F}_{D_{\max}}$ *for every* $d \geq D_{\max}$, *and*

$$\bigcup_{d \geq n} \mathcal{F}_d \;=\; \mathcal{F}_{D_{\max}} \;=\; \operatorname{cl} \mathcal{P},$$

a closed set. So $h_d(V) = h_{D_{\max}}(V) = \operatorname{dist}(V, \operatorname{cl} \mathcal{P})$ *for every* V *and every* $d \geq D_{\max}$; $\mathrm{rk}_{\mathrm{mem}}(V) \leq D_{\max}$ *for every* $V \in \operatorname{cl} \mathcal{P}$; *and* $\mathrm{rk}_{\mathrm{mem}}(V) = \infty$ *for every* $V \notin \operatorname{cl} \mathcal{P}$.

Proof. (a) Let $H_k \subseteq [n]^{k+1}$ be the set of positive-probability histories $(x_0, \ldots, x_k)$ of a realizing process, $N_k = |H_k| \leq n^{k+1}$, and take the hidden space $[D] = \bigsqcup_{k=0}^{m} H_k$, a disjoint union of layers. The hidden chain occupies layer k at time t_k: the initial law places mass $\mathbb{P}(X_{t_0} = x_0)$ on each $(x_0) \in H_0$; the step matrix M_{k+1} sends $h = (x_0, \ldots, x_k) \in H_k$ to $h \cdot b = (x_0, \ldots, x_k, b) \in H_{k+1}$ with probability $\mathbb{P}(X_{t_{k+1}} = b \mid X_{t_0}{=}x_0, \ldots, X_{t_k}{=}x_k)$, a well-defined conditional since h has positive probability, and sends every state outside layer k to an arbitrary fixed state of layer $k{+}1$ (these sources carry zero mass at time t_k, so the choice is immaterial); columns are stochastic by construction. The labeling φ sends a history to its last symbol on each layer. The history process $(X_{t_0}, \ldots, X_{t_k})_k$ is Markov with these steps, its law restricted to the layers reproduces the path law of (X_{t_i}), the observed block masses at time t_k are $\mathbb{P}(X_{t_k} = a) > 0$, and the block conditionals (7) are exactly $\mathbb{P}(X_{t_j} = b \mid X_{t_i} = a) = V(j \leftarrow i)_{ba}$. Hence $V = \kappa(\theta)$ with $\theta \in \Theta_D^+$ in $\mathfrak{D}^{\mathrm{inhom}}$. The lift also homogenizes: because the layers are disjoint and the chain occupies layer k at time t_k, the single step matrix M whose column at $h \in H_k$ $(k < m)$ is the column of M_{k+1} at h, and whose column at $h \in H_m$ is arbitrary, implements every step: a state's column acts on positive mass only when its layer is occupied, and each layer is occupied at exactly one time. Hence $V = \kappa(\theta)$ with $\theta \in \Theta_D^+$ in $\mathfrak{D}^{\mathrm{hom}}$ as well.

(b) For $\theta \in \Theta_d^+$ the compressed family $\kappa(\theta)$ is process-induced by the observed process $\varphi(X_{t_i})$, whose one-time marginals are the positive block masses; elements of the closure are limits of such families.

(c) By Proposition 4.4 the sequence $h_d(V)$ is nonincreasing, so $\lim_d h_d(V) = \inf_d \operatorname{dist}(V, \mathcal{F}_d) = \operatorname{dist}\big(V, \bigcup_d \mathcal{F}_d\big) = \operatorname{dist}\big(V, \operatorname{cl} \bigcup_d \mathcal{F}_d\big)$, which vanishes exactly when $V \in \operatorname{cl} \bigcup_d \mathcal{F}_d$. Part (a) gives $\mathcal{P} \subseteq \bigcup_d \mathcal{F}_d$ and part (b) gives $\mathcal{F}_d \subseteq \operatorname{cl} \mathcal{P}$ for every d; taking unions and closures, $\operatorname{cl} \bigcup_d \mathcal{F}_d = \operatorname{cl} \mathcal{P}$.

(d) By (a) and the exact inclusions $\kappa(\Theta_d^+) \subseteq \kappa(\Theta_{d+1}^+)$ of Proposition 4.4, $\mathcal{P} \subseteq \kappa(\Theta_{D_{\max}}^+) \subseteq \mathcal{F}_{D_{\max}}$. The set $\mathcal{F}_{D_{\max}}$ is closed, so $\operatorname{cl} \mathcal{P} \subseteq \mathcal{F}_{D_{\max}}$. By (b), $\mathcal{F}_d \subseteq \operatorname{cl} \mathcal{P} \subseteq \mathcal{F}_{D_{\max}}$ for every d, and with the nesting $\mathcal{F}_{D_{\max}} \subseteq \mathcal{F}_d$ for $d \geq D_{\max}$ this gives $\mathcal{F}_d = \mathcal{F}_{D_{\max}}$ for $d \geq D_{\max}$ and $\bigcup_d \mathcal{F}_d = \mathcal{F}_{D_{\max}} = \operatorname{cl} \mathcal{P}$. The statements about h_d and $\mathrm{rk}_{\mathrm{mem}}$ restate these equalities through Definitions 4.2 and 5.1. □

Two remarks delimit what the theorem proves. First, the dimension D of part (a) depends on the realizing process through the history counts N_k, while $D_{\max}$ depends on n and m only; by the uniform bound, (d) is a statement about $\operatorname{cl} \mathcal{P}$ and not about particular processes. Second, $D_{\max}$ is exponential in the number of grid steps and crude as a rank bound: for $n = 2$ and $m = 4$, $D_{\max} = 62$, while a lumped 4-state chain on that grid can have memory rank at most 3 (Example 8.1). Sharpening $D_{\max}$ toward the positive-realization bounds of [1, 31] is Open Problem (iii).

Remark 5.8 (Homogenization and the hidden clock). Both classes terminate at the same dimension because the history lift homogenizes: the layer index of a history is a clock, so the homogeneous realization of part (a) contains a copy of the time variable, and the homogeneity of its single step matrix carries no dynamical content beyond that of the inhomogeneous realization. The classes differ in the *minimal* dimension: $\mathrm{rk}_{\mathrm{mem}}{}^{\mathrm{hom}} \geq \mathrm{rk}_{\mathrm{mem}}{}^{\mathrm{inhom}}$ pointwise, by the inclusion of Proposition 7.1, and the inequality is strict for a flat family with strictly positive kernels and unequal one-step matrices, which is process-induced with $\mathrm{rk}_{\mathrm{mem}}{}^{\mathrm{inhom}} = n$

while $h_n^{\mathrm{hom}} > 0$ by Proposition 7.1. A homogeneous lift of such a family exists by part (a), at a dimension large enough to encode the clock. The class dependence of Section 7 is thus a statement about the profile at a fixed dimension, and not about eventual liftability, which holds in both classes by $D_{\max}$.

Corollary 5.9 (The asymptotic liftability class is empty)**.** *Define the* asymptotic liftability class *of the declared dilation class as*

$$\mathcal{A} \;:=\; \mathrm{cl}\,\mathcal{P} \;\setminus\; \bigcup_{d\ge n} \mathcal{F}_d \;=\; \big\{\, V :\; h_d(V) \downarrow 0 \text{ but } h_d(V) > 0 \text{ for every finite } d \,\big\},$$

the families asymptotically liftable but of infinite memory rank. Then $\mathcal{A} = \emptyset$ in either class. Equivalently, for every $V \in \mathcal{V}$ the three statements $\mathrm{rk}_{\mathrm{mem}}(V) < \infty$, $\lim_d h_d(V) = 0$, and $V \in \mathcal{F}_{D_{\max}}$ are equivalent, and $\mathrm{rk}_{\mathrm{mem}}(V) \le D_{\max}$ whenever they hold.

Proof. Theorem 5.7(d): $\mathrm{cl}\,\mathcal{P} = \bigcup_d \mathcal{F}_d$. □

5.3 The joint-law defect

Lemma 5.4 extracts from a model family one consequence of the existence of a joint law, the consistency of its one-time marginals. The existence of the joint law itself is a stronger condition, and it is also a linear program.

Definition 5.10 (Joint-law defect)**.** Let $\mathcal{L}([n]^{m+1})$ be the set of joint laws q of $(X_{t_0}, \dots, X_{t_m})$ on $[n]$, a simplex of dimension $n^{m+1} - 1$, and for q in it write $q_i(a) = q(X_{t_i} = a)$ and $q_{ij}(a,b) = q(X_{t_i} = a, X_{t_j} = b)$. The *joint-law defect* of $V \in \mathcal{V}$ is

$$\kappa_J(V) \;=\; \min_{q \in \mathcal{L}([n]^{m+1})} \; \max_{i<j} \; \max_{a,b\in[n]} \big|\, q_{ij}(a,b) - V(j \leftarrow i)_{ba}\, q_i(a) \,\big|.$$

Each term inside the maximum is affine in q, so $\kappa_J(V)$ is the value of a linear program in epigraph form with $n^{m+1} + 1$ variables; the minimum is attained, and $V \mapsto \kappa_J(V)$ is continuous.

Theorem 5.11 (Joint-law obstruction)**.** *The following hold in either dilation class.*

(a) $\kappa_J \equiv 0$ *on* $\mathcal{P}$ *and on every* $\mathcal{F}_d$*; in particular* $\kappa_J \equiv 0$ *on* $\mathrm{cl}\,\mathcal{P} = \mathcal{F}_{D_{\max}}$*.*

(b) κ_J *is Lipschitz with constant* $C' = \big(\min_{i<j,\,a} p_{ij}\,\omega_i(a)\big)^{-1/2}$*:* $|\kappa_J(V) - \kappa_J(W)| \le C'\,\mathrm{dist}(V,W)$*. So for every* $V \in \mathcal{V}$ *and every* $d \ge n$*,*

$$h_d(V) \;\ge\; \frac{\kappa_J(V)}{C'},$$

and $\kappa_J(V) > 0$ *implies* $\mathrm{rk}_{\mathrm{mem}}(V) = \infty$*.*

(c) $\kappa_c(V) \le n^2\,\kappa_J(V)$ *for every* V*. So* $\kappa_J(V) = 0$ *implies* $\kappa_c(V) = 0$*.*

(d) $V \in \mathcal{P}$ *if and only if the linear program of Definition 5.10 attains the value* 0 *at a joint law* q *with* $q_i(a) > 0$ *for all* i, a*.*

Proof. (a) For $V \in \mathcal{P}$ the law of a realizing process is feasible with all terms zero, since $q_{ij}(a,b) = \mathbb{P}(X_{t_j} = b \mid X_{t_i} = a)\,q_i(a)$. For $\kappa(\theta)$ with $\theta \in \Theta_d^+$ the law of the observed process $\varphi(X_{t_i})$ serves; continuity extends the vanishing to $\mathcal{F}_d$, and Theorem 5.7(d) identifies $\mathrm{cl}\,\mathcal{P}$ with $\mathcal{F}_{D_{\max}}$.

(b) Fix q. For each (i,j,a,b) the difference of the two terms for V and W is $\big|V(j \leftarrow i)_{ba} - W(j \leftarrow i)_{ba}\big|\,q_i(a) \le |X_{ba}|$ with $X = (V - W)(j \leftarrow i)$, and (4) gives $p_{ij}\,\omega_i(a)\,X_{ba}^2 \le \mathrm{dist}(V,W)^2$, so each term moves by at most $C'\,\mathrm{dist}(V,W)$; the maximum over indices and the minimum over

q preserve the bound. The inequality for h_d follows by evaluating at a nearest point of $\mathcal{F}_d$, where κ_J vanishes by (a).

(c) Let q attain $\kappa_J(V)$ and put $\mu_i = q_i$. For each $i < j$ and b, $\big(V(j \leftarrow i)\mu_i - \mu_j\big)_b = \sum_a \big(V(j \leftarrow i)_{ba}\, q_i(a) - q_{ij}(a,b)\big)$, since $\sum_a q_{ij}(a,b) = q_j(b)$; so $|\cdot| \le n\,\kappa_J(V)$ and $\|V(j \leftarrow i)\mu_i - \mu_j\|_1 \le n^2\kappa_J(V)$. Minimizing over μ gives the claim.

(d) If q attains 0 with positive one-time marginals, then $V(j \leftarrow i)_{ba} = q_{ij}(a,b)/q_i(a)$ is the conditional law of the process with law q, so $V \in \mathcal{P}$; the converse is (a). □

The two obstructions are ordered as detectors and not as floors. By (c), κ_J detects every family that κ_c detects, and it detects more: the converse of (c) fails, and Example 8.3 exhibits a family with $\kappa_c = 0 < \kappa_J$. Marginal consistency is a pairwise condition on one-time laws, while κ_J tests the existence of a joint law, which the finite marginal problem of Vorob'ev and Kellerer shows to be a strictly stronger requirement in general [32, 19]. As floors on the profile the two constants differ, $C_* = \sqrt{n}\,C'$, and on a given family either floor may be the larger, so the larger one applies; in Example 8.2 the marginal-consistency floor is the larger, and in Example 8.3 only the joint-law floor is positive. The condition in (d) is decidable by a second linear program that maximizes the minimal one-time marginal over the optimal face of the first.

Remark 5.12 (What the memory rank measures). The memory rank is the least hidden dimension, in the closed model class, at which the two-time conditionals of V are reproduced. A realization of the full path law of a process on the grid as a function of a Markov chain on D states, a lumped chain, is a lift in the sense of Definition 4.1, so for $V \in \mathcal{P}$ the memory rank is at most the minimal order of such a realization of the path law of any process inducing V; a hidden Markov model with stochastic emissions on D states becomes such a realization on the state–symbol pairs, at most nD states, so the memory rank is at most n times the minimal order of a hidden Markov model realizing the path law. The inequalities can be strict, because the rank constrains two-time conditionals only and a family in $\mathcal{P}$ may be induced by several processes. No inequality between the memory rank and the Hankel rank of a path law, or the nonnegative rank of its two-time joint matrices, is claimed in either direction; Open Problem (iii) asks for such comparisons. The converse of Theorem 5.11(a), from $\kappa_J(V) = 0$ to $V \in \mathcal{F}_{D_{\max}}$, holds when some optimal law of the linear program has positive one-time marginals, by (d); when every optimal law has a vanishing marginal we do not know whether it holds.

6 First-order conditions at the projection

This section records the first-order conditions that a nearest point must satisfy. They are the standard necessary conditions of variational analysis for the projection onto a closed semialgebraic set, stated here because the companion papers use them; Remark 6.1 records the one model-specific fact, the fibers of the compression map.

Fix d and $V \notin \mathcal{F}_d$, and let $W^* \in \mathcal{F}_d$ attain $h_d(V)$ (Proposition 4.3). Because $\mathcal{F}_d$ is closed, the residual $V - W^*$, the model mismatch at the nearest point, belongs to the regular normal cone $\widehat{N}_{\mathcal{F}_d}(W^*)$, equivalently

$$\langle V - W^*,\, Z - W^* \rangle \;\le\; o(\|Z - W^*\|) \qquad \text{as } Z \to W^*,\ Z \in \mathcal{F}_d, \tag{10}$$

see [26, Ch. 6]. The residual is a difference of families; it is not the deficit Δ_V, and the two are related only through Proposition 5.3 at $d = n$. Refinements follow from the stratification of Proposition 4.5. Let $S \subseteq \mathcal{F}_d$ be the smooth stratum containing W^*. Since $S \subseteq \mathcal{F}_d$, normal cones reverse the inclusion, $\widehat{N}_{\mathcal{F}_d}(W^*) \subseteq N_S(W^*)$, so (10) implies the orthogonality

$$V - W^* \;\perp\; T_S(W^*), \tag{11}$$

and the residual decomposes along $T_S(W^*) \oplus N_S(W^*)$ with vanishing tangential part: at the nearest point the residual is normal to the stratum. When W^* lies on a boundary stratum, (10) remains the necessary condition, and its cone form is the general statement: the residual lies in the polar of the contingent tangent cone, $V - W^* \in T_{\mathcal{F}_d}(W^*)^\circ$. No Moreau decomposition against $T_{\mathcal{F}_d}(W^*)$ is asserted, because that cone need not be convex; the decomposition applies to the closed convex Clarke tangent cone $T^C_{\mathcal{F}_d}(W^*) \subseteq T_{\mathcal{F}_d}(W^*)$ and its polar [26, Ch. 6].

Two standard facts complete the local picture. First, the set of V with non-unique nearest point in the closed set $\mathcal{F}_d$ has Lebesgue measure zero relative to the affine hull of $\mathcal{V}$, since the squared distance is locally semiconcave and hence differentiable almost everywhere, and differentiability forces uniqueness [26, Ch. 6]; "generic" here means Lebesgue almost everywhere on that affine hull, not open-dense and not Zariski-generic. Second, for generic ambient data, the number of isolated smooth critical points of the squared-distance function on the smooth locus of the Zariski closure of a stratum is bounded by the Euclidean distance (ED) degree of that closure [11]; the count carries the standard qualifications, since singularities of the closure, critical points at infinity, real-boundary constraints invisible to the Zariski closure, and nongeneric targets with positive-dimensional critical loci all fall outside it. Computing the ED degrees of the strata of $\mathcal{F}_d$, under those qualifications, is Open Problem (iv).

Remark 6.1 (Fibers of the compression). For $\theta = (\pi, (M_k), \varphi) \in \Theta^+_{d,\varphi}$ let $g = (g_0, \dots, g_m)$ be invertible $d \times d$ matrices, block-diagonal with respect to the blocks $\varphi^{-1}(a)$ and mass-preserving on each block, $\mathbf{1}_a^\top g_i = \mathbf{1}_a^\top$; in $\mathfrak{D}^{\mathrm{hom}}$ take all g_i equal. Then $\theta^g = (g_0\pi, (g_k M_k g_{k-1}^{-1}), \varphi)$ has the same hidden block masses and the same block conditionals as θ, because g_i commutes with $\mathrm{diag}(\mathbf{1}_a)$ and preserves $\mathbf{1}_a^\top$, so $\kappa(\theta^g) = \kappa(\theta)$ whenever θ^g is a lift, which holds for g near the identity when the parameters of θ are strictly positive. The group has dimension $\sum_a d_a(d_a - 1)$ in $\mathfrak{D}^{\mathrm{hom}}$ and $(m+1)\sum_a d_a(d_a-1)$ in $\mathfrak{D}^{\mathrm{inhom}}$, where $d_a = |\varphi^{-1}(a)|$, and its orbit through θ has the dimension of the group minus the dimension of the stabilizer of θ. At a point whose stabilizer is discrete the fibers of κ thus have dimension at least the group dimension, and the rank of the differential of κ is at most the dimension of $\Theta^+_{d,\varphi}$ minus the group dimension, and at most the dimension $\binom{m+1}{2} n(n-1)$ of the ambient family space; at $d = n$ the initial law does not enter the compression, which removes $n - 1$ further dimensions. We do not prove that the stabilizer is discrete for generic θ. So at a nearest point $W^* = \kappa(\theta)$ with θ strictly positive, the image of the differential of κ at θ is a linear subspace of the contingent cone $T_{\mathcal{F}_d}(W^*)$, of the dimension just bounded, and the cone form of (10) implies orthogonality of the residual to that image; the ED-degree count of the previous paragraph concerns the Zariski closures of the strata.

7 Class dependence

Proposition 7.1. *For every $d \geq n$, $\mathcal{F}_d^{\mathrm{hom}} \subseteq \mathcal{F}_d^{\mathrm{inhom}}$ and hence $h_d^{\mathrm{inhom}} \leq h_d^{\mathrm{hom}}$ pointwise. The inequality is strict at the bottom of the profile on explicit families: there exist V with $h_n^{\mathrm{inhom}}(V) = 0 < h_n^{\mathrm{hom}}(V)$, and the gap admits an explicit lower bound.*

Proof. The inclusion of classes gives the inclusion of model sets and the inequality. For strictness take $d = n$, so labelings are bijections and, as in Example 5.2, $\mathcal{F}_n^{\mathrm{inhom}} = \mathcal{F}\mathrm{lat}$ while every $W \in \mathcal{F}_n^{\mathrm{hom}}$ satisfies $W(1 \leftarrow 0) = W(2 \leftarrow 1)$, a closed condition preserved under the closure. Choose V flat with one-step kernels $A = V(1 \leftarrow 0) \neq B = V(2 \leftarrow 1)$ and $V(j \leftarrow i)$ the corresponding products. Then $h_n^{\mathrm{inhom}}(V) = 0$. For any $W \in \mathcal{F}_n^{\mathrm{hom}}$ with common one-step kernel M, the two weighted summands for the pairs $(0,1)$ and $(1,2)$ in (4) carry different state weights ω_0 and ω_1, so we pass to a common Frobenius lower bound: with $c := \min_{i,a} \omega_i(a) > 0$

and $\|\cdot\|_F$ the Frobenius norm,

$$\operatorname{dist}(V,W)^2 \;\geq\; c\,\min(p_{01},p_{12})\big(\|A-M\|_F^2+\|B-M\|_F^2\big) \;\geq\; \tfrac{c}{2}\,\min(p_{01},p_{12})\,\|A-B\|_F^2,$$

by the parallelogram identity in the Frobenius norm, the minimum over M being attained at the stochastic matrix $(A+B)/2$. So $h_n^{\mathrm{hom}}(V) \geq \big(\frac{c}{2}\min(p_{01},p_{12})\big)^{1/2}\|A-B\|_F > 0$. (When the state weights coincide, $\omega_0=\omega_1=\omega$, the same chain runs directly in the weighted norm $\|\cdot\|_\omega$ without the Frobenius passage.) □

The strictness is a statement at the dimension-matched level $d=n$. At larger d the homogeneous class can encode time in a hidden clock, as the history lift of Theorem 5.7(a) does, so the family of the proof lies in $\mathcal{F}_D^{\mathrm{hom}}$ for some $D \leq D_{\max}$ (Remark 5.8); the class exchanges hidden dimension for time-inhomogeneity, and the profile at a fixed d records the rate. The phenomenon is the temporal analog of the dependence of lift size on the lifting cone in extension-complexity theory [33]: the same observed object has different minimal explanations, and different distances to explanation, in different classes. Example 8.4 computes the gap in closed form on the smallest instance.

8 Exact examples

Each example in this section is a statement about a specific family, proved from the definitions by hand. Examples 8.1 and 8.5 hold for every n and every grid. Examples 8.2 to 8.4 use $n=2$, the grid $\mathbb{T}=\{t_0,t_1,t_2\}$, and the weights $p_{ij}=1$ and $\omega_i=\mathbf{1}$ in (4), so the metric is the stacked Frobenius norm, $C_*=\sqrt{2}$ in Theorem 5.6, and $C'=1$ in Theorem 5.11. Write e_1,e_2 for the standard basis of $\mathbb{R}^2$, I for the identity, $S=\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ for the swap, $R_a=e_a\mathbf{1}^\top$ for the reset to state a, and $J=\frac{1}{2}\mathbf{1}\mathbf{1}^\top$ for the uniform reset. States are labeled 1 and 2.

Example 8.1 (Memory rank below the generating dimension). Let $\theta \in \Theta_d^+$ be a lift in either class, on any grid, and let $\theta' \in \Theta_{d+1}^+$ be its split at a hidden state u by the construction of Proposition 4.4. The family $V=\kappa(\theta')$ is the compression of a lumped $(d{+}1)$-state chain, and $\kappa(\theta')=\kappa(\theta)$, so $V \in \kappa(\Theta_d^+)$, $h_d(V)=0$, and $\mathrm{rk}_{\mathrm{mem}}(V) \leq d$. So the memory rank of a family is at most the generating dimension of any lumped chain inducing it, and the inequality is strict for every family induced by a split lift. On $n=2$, split the singleton block of a 3-state lift with labeling $\varphi=(1,2,2)$: the result is a 4-state chain observed through the two-and-two labeling $(1,1,2,2)$, and its observed family has memory rank at most 3.

Example 8.2 (A family separated from every model set). Let $V(1\leftarrow 0)=R_1$, $V(2\leftarrow 1)=R_1$, and $V(2\leftarrow 0)=R_2$. Then $\kappa_c(V)=1$ and $\kappa_J(V)=\frac{1}{4}$, so $h_d(V) \geq 1/\sqrt{2}$ for every $d\geq 2$ in either class, and $\mathrm{rk}_{\mathrm{mem}}(V)=\infty$.

For κ_c: $R_a\mu=e_a$ for every probability vector μ, so the three residuals of Definition 5.5 are $e_1-\mu_1$, $e_1-\mu_2$, and $e_2-\mu_2$, and $\|e_1-\mu_2\|_1+\|e_2-\mu_2\|_1 \geq \|e_1-e_2\|_1=2$ makes the maximum at least 1 for every choice of μ; the choice $\mu_1=e_1$, $\mu_2=\frac{1}{2}(e_1+e_2)$ gives the residual norms $0,1,1$, so $\kappa_c(V)=1$.

For κ_J: let q be a joint law with maximal deviation t in Definition 5.10. The pair $(0,1)$ with $b=2$ gives $q_{01}(a,2) \leq t$ for each a, since $V(1\leftarrow 0)_{2a}=0$, so $q(X_{t_1}=2) \leq 2t$; the pair $(1,2)$ with $b=2$ gives $q(X_{t_2}=2)\leq 2t$ in the same way; and the pair $(0,2)$ with $b=2$ gives $q_{02}(a,2) \geq q_0(a)-t$ for each a, since $V(2\leftarrow 0)_{2a}=1$, so $q(X_{t_2}=2)\geq 1-2t$. Thus $1-2t\leq 2t$ and $t\geq\frac{1}{4}$. The law q uniform on the four paths $(1,1,1)$, $(1,1,2)$, $(1,2,2)$, $(2,2,1)$ has every deviation equal to $\frac{1}{4}$, so $\kappa_J(V)=\frac{1}{4}$. Here $\kappa_J=\kappa_c/n^2$, so the inequality of Theorem 5.11(c) is attained. The two floors are $\kappa_c/C_*=1/\sqrt{2}$ and $\kappa_J/C'=\frac{1}{4}$, and the first applies.

Example 8.3 (Consistent marginals without a joint law). Let $V(1\leftarrow 0)=V(2\leftarrow 1)=I$ and $V(2\leftarrow 0)=S$. The uniform laws $\mu_0=\mu_1=\mu_2=\frac{1}{2}\mathbf{1}$ satisfy (8), so $\kappa_c(V)=0$; the family is

not flat, since $\Delta_V(2,0,1) = S - I$; and $\kappa_J(V) = \frac{1}{6}$, so $h_d(V) \geq \frac{1}{6}$ for every $d \geq 2$ in either class and $\mathrm{rk}_{\mathrm{mem}}(V) = \infty$.

For the lower bound, let q be a joint law with maximal deviation t. The pair $(0,1)$ gives $q_{01}(a,b) \leq t$ for $b \neq a$, since $I_{ba} = 0$, so $q(X_{t_0} \neq X_{t_1}) \leq 2t$; the pair $(1,2)$ gives $q(X_{t_1} \neq X_{t_2}) \leq 2t$; and the pair $(0,2)$ gives $q_{02}(a,a) \leq t$, since $S_{aa} = 0$, so $q(X_{t_0} = X_{t_2}) \leq 2t$. The event $\{X_{t_0} = X_{t_1}\} \cap \{X_{t_1} = X_{t_2}\}$ is contained in $\{X_{t_0} = X_{t_2}\}$, so $1 \leq q(X_{t_0} \neq X_{t_1}) + q(X_{t_1} \neq X_{t_2}) + q(X_{t_0} = X_{t_2}) \leq 6t$. The law q uniform on the six paths other than $(1,2,1)$ and $(2,1,2)$ has every deviation equal to $\frac{1}{6}$, so $\kappa_J(V) = \frac{1}{6}$. Each kernel of V is the two-time conditional of a process with uniform marginals, and the three pairwise laws they determine are consistent with one another; no law on $[2]^3$ has all three as its two-time marginals. This is the smallest instance of the marginal problem of Vorob'ev and Kellerer [32, 19] in the setting of this paper, and it separates the two obstructions: $\kappa_c(V) = 0$ and $\kappa_J(V) = \frac{1}{6}$.

Example 8.4 (The class gap in closed form). Let $V(1 \leftarrow 0) = I$, $V(2 \leftarrow 1) = S$, and $V(2 \leftarrow 0) = SI = S$. The family is flat with unequal one-step kernels, and it is induced by the process with X_{t_0} uniform, $X_{t_1} = X_{t_0}$, and $X_{t_2} = 3 - X_{t_1}$, so $h_2^{\mathrm{inhom}}(V) = 0$, $\mathrm{rk}_{\mathrm{mem}}{}^{\mathrm{inhom}}(V) = 2$, and $\mathrm{rk}_{\mathrm{mem}}{}^{\mathrm{hom}}(V) \leq 6$ by Theorem 5.7(a), since $N_0 = N_1 = N_2 = 2$. In the homogeneous class,

$$h_2^{\mathrm{hom}}(V) \;=\; \sqrt{3},$$

against the lower bound $\sqrt{2}$ that the proof of Proposition 7.1 gives.

At $d = n = 2$ every labeling is a bijection, a homogeneous lift with step matrix M has compression (M, M, M^2), and $\mathcal{F}_2^{\mathrm{hom}} = \{(M, M, M^2) : M \in \mathcal{S}_2\}$: the set on the right is the continuous image of the compact $\mathcal{S}_2$, so it is closed, and every M is a limit of strictly positive M', whose lifts with $\pi = \frac{1}{2}\mathbf{1}$ have positive block masses. So $h_2^{\mathrm{hom}}(V)^2 = \min_{M \in \mathcal{S}_2} f(M)$ with $f(M) = \|I - M\|_F^2 + \|S - M\|_F^2 + \|S - M^2\|_F^2$. The parallelogram identity gives $\|I - M\|_F^2 + \|S - M\|_F^2 = \frac{1}{2}\|I - S\|_F^2 + 2\|M - J\|_F^2 = 2 + 2\|M - J\|_F^2$. Write $M = J + E$ with $E = \left(\begin{smallmatrix} x & y \\ -x & -y \end{smallmatrix}\right)$ and $x, y \in [-\frac{1}{2}, \frac{1}{2}]$. Then $JE = 0$, $E^2 = (x - y)E$, and $M^2 = J + EJ + (x - y)E$. On the matrices with zero column sums use the coordinates $(x, y) \mapsto \left(\begin{smallmatrix} x & y \\ -x & -y \end{smallmatrix}\right)$, in which the squared Frobenius norm is $2(x^2 + y^2)$; then $S - J$ has coordinates $(-\frac{1}{2}, \frac{1}{2})$, EJ has coordinates $\frac{1}{2}(x + y)(1, 1)$, and E has coordinates (x, y), so $S - M^2$ has coordinates $(-\frac{1}{2} - p,\ \frac{1}{2} - r)$ with $p = \frac{1}{2}(x + y) + xu$, $r = \frac{1}{2}(x + y) + yu$, and $u = x - y$. Since $p - r = u^2$,

$$\|S - M^2\|_F^2 \;=\; 2\big[(\tfrac{1}{2} + p)^2 + (\tfrac{1}{2} - r)^2\big] \;=\; 1 + 2u^2 + 2p^2 + 2r^2,$$

and with $\|M - J\|_F^2 = 2(x^2 + y^2)$,

$$f(M) \;=\; 3 + 4(x^2 + y^2) + 2u^2 + 2p^2 + 2r^2 \;\geq\; 3,$$

with equality exactly at $E = 0$, that is, at $M = J$. So $h_2^{\mathrm{hom}}(V) = \sqrt{3}$, attained at the uniform reset lift, whose compression is (J, J, J), and the bound $\sqrt{2}$ of Proposition 7.1 is not attained on this family.

Example 8.5 (The gauge path in closed form). Let $V \in \mathcal{V}$ on any grid, let $w \in \mathbb{R}^n_{>0}$ be a probability vector, and let $g = g^{(\varepsilon)}$ be the re-dressing of Theorem 3.3. Since $\mathbf{1}^\top \Delta_V = 0$, $g_t \Delta_V(t, s, r) = \varepsilon_t \Delta_V(t, s, r)$, and the inverse displayed in the proof of the theorem gives $\Delta_V(t, s, r) g_s^{-1} = \varepsilon_s^{-1} \Delta_V(t, s, r)\big(I - (1 - \varepsilon_s) w \mathbf{1}^\top\big)$. By Lemma 3.2,

$$\begin{aligned} \Delta_{V^g}(t, s, r) \;&=\; \varepsilon^{\,t-s}\, \Delta_V(t, s, r)\big(I - (1 - \varepsilon^{\,s+1})\, w \mathbf{1}^\top\big), \\ \|\Delta_{V^g}(t, s, r)\|_F \;&\leq\; (1 + \sqrt{n})\, \varepsilon^{\,t-s}\, \|\Delta_V(t, s, r)\|_F, \end{aligned}$$

the second by $\|I - c\, w \mathbf{1}^\top\|_{\mathrm{op}} \leq 1 + c\|w\|_2 \|\mathbf{1}\|_2 \leq 1 + \sqrt{n}$ for $c \in [0, 1]$. So the deficit of every triple decays at the rate $\varepsilon^{\,t-s}$, the adjacent triples $t - s = 2$ slowest, and the transformed family approaches the reset form at the rate ε: by the display in the proof of Theorem 3.3,

$V^g(j \leftarrow i) - w\mathbf{1}^\top = \rho_{ij}\big(A - (1-\varepsilon_i)Aw\mathbf{1}^\top\big) - \varepsilon_j w\mathbf{1}^\top$ with $A = V(j \leftarrow i)$, whose Frobenius norm is at most $2\sqrt{n}\,\varepsilon^{j-i} + \sqrt{n}\,\varepsilon^{j+1}$, since $\|A\|_F \le \sqrt{n}$ and $\|Aw\mathbf{1}^\top\|_F \le \sqrt{n}$ for column-stochastic A. The distance to $\mathcal{F}$lat decays at the rate ε^2, the rate of the slowest deficit triples. Write $V^g(k \leftarrow k-1) = w\mathbf{1}^\top + \varepsilon C_k$ with $C_k = A_k - (1-\varepsilon^k)A_k w\mathbf{1}^\top - \varepsilon^k w\mathbf{1}^\top$ and $A_k = V(k \leftarrow k-1)$; then $\mathbf{1}^\top C_k = 0$ and $C_k w = \varepsilon^k(A_k w - w)$, so every product of consecutive one-step kernels of V^g equals $w\mathbf{1}^\top + O(\varepsilon^2)$, and so does every $V^g(j \leftarrow i)$ with $j - i \ge 2$ by the display above. The flat family generated by the one-step kernels of V^g is thus within $O(\varepsilon^2)$ of V^g, and $\operatorname{dist}(V^g, \mathcal{F}\mathrm{lat}) = O(\varepsilon^2)$. Along this path the conditioning $\|g_i\|_{\mathrm{op}}\|g_i^{-1}\|_{\mathrm{op}}$ is of order $\varepsilon^{-(i+1)}$, and equals it for uniform w; so with K a bound on the largest conditioning, attained at $i = m$, the surrogate F_K of Remark 3.6 satisfies $F_K(V) = O(K^{-2/(m+1)})$ as $K \to \infty$, with constants depending on n, m, and V.

9 Open problems

(i) *Selection bridge.* The selection functionals of [30] extract predictive values from a family without a generator by reweighting proposed paths against a surrogate. The problem is one of design: specify a triple consisting of a selection functional, a class of surrogates, and a readout such that, for every process-induced V and every nearest point W^* of V in $\mathcal{F}_d$, the readout of V and the readout of W^* differ by an error controlled by $h_d(V)$. On the zero stratum V is itself a nearest point, so the clause of equality there is automatic and carries no content; the content is the bound off the stratum, quantified over all nearest points because the projection need not be unique (Section 6). The same problem is posed, in the same design form, from the estimation side in [30], and it is open in both directions.

(ii) *Natural contractions.* Classify the contractions of the deficit Δ_V, functionals of Δ_V taking values in a normed space, that are natural with respect to lumpings and dilations and compatible with the coherence identity (5) and the curvature formula (6). The dimension-matched charge h_n is not a contraction of Δ_V; it is a distance to a model set, related to the deficit norms only through Proposition 5.3. A classification would determine which contractions, if any, are comparable to h_n with an exponent $\alpha = 1$.

(iii) *Sharp rank bounds and comparisons.* Replace $D_{\max}$ in Theorem 5.7(d) by a bound in the spirit of positive-realization theory [1, 31]; relate the memory rank of a process-induced family to the minimal order of hidden Markov realizations of the path laws that induce it, to the Hankel ranks of those laws, and to the nonnegative ranks of their two-time joint matrices [9, 21], comparing quantities defined on the same data, as Remark 5.12 requires; and characterize the families on the boundary strata of $\mathcal{F}_d$.

(iv) *ED degrees.* Compute or bound the Euclidean distance degrees [11] of the Zariski closures of the strata of $\mathcal{F}_d$ as functions of (n, d, m) and the class, under the standard qualifications of Section 6, thereby bounding the number of smooth critical points of the squared distance on those strata.

(v) *Behavior under lumpings.* For $V \in \mathcal{P}$ induced by a process X and a lumping $\Lambda : [n] \to [n']$ of the observed states, let ΛV denote the family induced by the process $\Lambda(X)$, when its one-time marginals are positive; the definition depends on the inducing process, and the problem is posed on $\mathcal{P}$ for that reason. Decompose $h_{n'}(\Lambda V)$ into a contractive image of the fine charge $h_n(V)$ and a production term with an identified zero set. The raw deficit Δ_V is a quadratic polynomial in the family by (2), so at that level the decomposition is a question about rational functions of the fine kernels and marginals; the corresponding decomposition of the metric charge is open.

(vi) *Continuous grids.* Formulate the profile for families over an interval, where $\mathcal{V}$ is infinite-dimensional and compactness of $\mathcal{F}_d$ must be argued in an appropriate topology. Proposition 4.11 makes the rank monotone under refinement of the grid, and $D_{\max}$ grows exponentially with the

number of grid steps, so the first question is whether the ranks along a refining sequence of grids stabilize, and at what dimension.

(vii) *Compactified classes.* Construct a compactified dilation class, lifts carrying boundary data such as conditional kernels on blocks of vanishing mass, whose compression image is exactly $\mathcal{F}_d$, restoring exact-realization language for the zero stratum (Remark 4.9). Example 4.10 shows what the boundary data must record: on a block whose mass has vanished, the limiting column of the compression is inherited from the direction of approach. Lemma 5.4, which passes marginal consistency to limits, is the template for the argument that the compactified image is exactly $\mathcal{F}_d$. Theorem 5.7(d) settles termination on the closure without a compactification, since $\bigcup_d \mathcal{F}_d$ is closed; the realization language remains open, and the question of Remark 4.9, whether for $d > n$ a family in the closure gap can admit strictly positive consistent one-time marginals, remains open as well.

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